

Multi Domain Data Fusion for Supply Chain Risk Prediction: A Comparative Analysis of SVM, MLP, and Random Forest Architectures

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Abstract—Global supply chain disruptions are high dimensional stochastic events that traditional univariate models fail to capture. This research proposes a heterogeneous data fusion framework synthesizing internal logistics logs with macroeconomic indicators from the New York Federal Reserve and localized meteorological data. We implement, tune, and debug three distinct architectures: Support Vector Machines (SVM), Multi Layer Perceptrons (MLP), and Random Forest Ensembles. Our results identify an “Accuracy Recall Paradox” in imbalanced logistics data. While the MLP and Random Forest achieved higher nominal accuracy, the Balanced SVM proved superior for industrial risk detection with a recall of 0.74.

I. INTRODUCTION

A. The Industrial Imperative: From Efficiency to Resilience

The shift in global logistics from a “Just in Time” to a “Just in Case” philosophy is not merely a trend but a survival imperative. The economic cost of supply chain blindness is staggering; a single week of blockage at a major maritime artery, such as the Suez Canal, can stall billions of dollars in global trade. This research is motivated by the need for “Prescriptive Resilience” or the ability of a system not only to predict that a delay will occur but to understand the multi domain stressors causing it. By moving beyond simple lead time averages, this project provides a blueprint for an early warning system that can assist supply chain managers in preemptively rerouting cargo or renegotiating delivery windows before a crisis peaks.

B. Theoretical Context: Stochastic vs. Systemic Risk

A core contribution of this introduction is the formalization of risk types within the logistics domain.

- **Systemic Risk:** Represented in this study by the Global Supply Chain Pressure Index (GSCPI). These are the macro tides of the global economy such as inflationary pressures, national labor shortages, and geopolitical shifts that increase the baseline friction of moving goods.
- **Stochastic Risk:** Represented by localized weather patterns. These are the micro waves or sudden, unpredictable events like a high wind gust or a flash flood that provide the final tip into a state of disruption.

The interplay between these two is the central focus of our feature engineering. We argue that a model failing to account for both is essentially half blind, capable of seeing the storm but not the engine trouble, or vice versa.

C. Bridging the Technological Gap

Current Enterprise Resource Planning (ERP) systems often struggle with Data Heterogeneity. Merging the structured, relational data of a shipping log with the unstructured, time series nature of meteorological reports and the indexical nature of Federal Reserve data presents a significant software engineering hurdle. This report details the construction of a Middleware Fusion Layer, a preprocessing pipeline designed to handle asynchronous timestamps and varying spatial resolutions. This layer acts as the bridge between raw, multi domain data and the high dimensional input required for advanced machine learning architectures like Support Vector Machines (SVM) and Multi Layer Perceptrons (MLP).

D. Key Methodological Contributions

This project makes three distinct contributions to the field of applied machine learning in supply chain analytics:

- 1) **Macro Regional Proxying:** A novel imputation strategy that solves the problem of Localized Data Vacuums by mapping port locations to host country climate profiles.
- 2) **Multimodal Decision Boundaries:** The implementation of non linear kernels (RBF) to identify complex safety margins.
- 3) **Cost Sensitive Objective Functions:** Tuning algorithms to penalize False Negatives (missed disruptions) more heavily than False Positives, aligning model performance with the actual financial risks faced by logistics firms.

E. Summary of Engineering Rigor

This study is grounded in the principles of reproducible data science. Every step from the standardization of Federal Reserve macroeconomic signals to the debugging of vanishing gradients in deep neural networks is documented to ensure that the proposed framework can be audited and scaled within

a production environment. By treating the predictive task as a software engineering problem as much as a statistical one, we provide a robust solution that addresses the systemic vulnerabilities of modern trade.

II. RELATED WORK

A. Geometric Margin Maximization: The SVM Paradigm

In the context of non linear risk detection, Support Vector Machines (SVM) have gained significant traction in recent years as a robust alternative to neural networks for smaller, high dimensional datasets.

The Kernel Trick: A recurring theme in supply chain literature is the use of the Radial Basis Function (RBF) kernel to project raw logistics data into a higher dimensional Hilbert space. This allows the model to identify a maximum margin hyperplane that can separate Disrupted shipments from Normal ones even when the decision boundary is not linearly separable in the original feature space.

Regularization and Robustness: Unlike deep learning models that are prone to overfitting on sparse supply chain logs, SVMs utilize a regularization parameter (C) to manage the trade off between margin width and classification error. Existing research suggests that this geometric approach is often more resilient to the noise found in meteorological data than standard regression techniques.

B. Data Imputation and the Sparsity Challenge

One of the most persistent hurdles in supply chain analytics is Data Incompleteness. Meteorological stations are not consistently located at every global port, leading to holes in the dataset.

Traditional Imputation: Historically, researchers have used Mean or Median imputation to fill these gaps. However, as noted in recent data engineering studies, this flattening of data can strip away the very volatility that predicts a disruption.

Proxy Logic vs. Constant Filling: More advanced literature suggests using Regional Proxies or utilizing the weather patterns of a nearby city or a national average to represent a specific node. This research builds upon this by formalizing a Macro Regional Proxy Mapping that ensures the model retains a signal for every row without resorting to synthetic Zero filling, which would falsely imply perfect weather conditions.

C. Macroeconomic Indices as Leading Indicators

The integration of Federal Reserve Data, specifically the Global Supply Chain Pressure Index (GSCPI), represents a relatively new frontier in logistics modeling.

From Lagging to Leading Indicators: Most logistics models utilize Lagging Indicators or data about what has already happened such as a yesterday's port delay.

The Systemic Signal: Modern economic research suggests that indices like the GSCPI serve as Leading Indicators, capturing systemic friction such as inflation, labor shortages, and energy costs months before they manifest as a localized port bottleneck.

D. The Accuracy Recall Paradox in Industrial Contexts

Finally, the Accuracy Recall Paradox is increasingly discussed in the context of Industrial Utility.

The Cost of Misclassification: In the shipping industry, the cost of a False Negative or failing to predict a port closure involves liquidated damages, missed contract penalties, and lost customer trust. The cost of a False Positive or rerouting a ship unnecessarily is high but often lower than a total systemic failure.

Optimization Shift: Related work in high stakes fields like supply chain analytics is beginning to prioritize F1 Score and Recall over nominal accuracy.

E. Research Synthesis

In summary, while the components of this study including macroeconomics, weather, and machine learning exist individually in the literature, their synthesis into a single Heterogeneous Fusion Model is the primary research gap.

III. OUR SOLUTION: DATA ENGINEERING ARCHITECTURE

A. The Challenge of Asynchronous Data Fusion

The primary engineering hurdle of this project was the synchronization of three fundamentally incompatible data domains. We faced a Temporal and Spatial Resolution Conflict where the data did not naturally align.

- **The Resolution Conflict:** Logistics logs provided daily granularity; however, the Federal Reserve's GSCPI is a monthly metric.
- **The Solution:** We implemented a Temporal Normalization Function. Instead of attempting to interpolate daily economic data, we generated a shared YearMonth key.
- **Impact:** This ensured the model was grounded in true macroeconomic states rather than guessed values.

B. Addressing the Localized Data Vacuum

Initial merges resulted in a 40% loss of data due to missing weather values for specific port nodes.

Solution: Macro Regional Proxy Mapping. We developed a dictionary that mapped every port to its host country and utilized national climate volatility averages like wind speed and precipitation as a proxy for the port's environment. While a national average is less precise than a port side sensor, it preserved the volatility signal without requiring the deletion of 40% of our logistics records.

```
# Author: Cian O'Mahony
# Mapping logistics ports to host countries to
# address sparsity
port_to_country = {
    'Los Angeles': 'United States of America',
    'Shanghai': 'China',
    'Rotterdam': 'Netherlands',
    'Busan': 'South Korea',
    'Dubai': 'United Arab Emirates'
}
merged_df['Country'] = merged_df['Destination_Port']
    .map(port_to_country)
```

```
# Calculating National Climate Profiles for
# Imputation
avg_weather = weather_df.groupby('country')[['gust_kph', 'precip_mm']].mean().reset_index()
final_dataset = pd.merge(merged_df, avg_weather,
    left_on='Country', right_on='country', how='left')
```

Listing 1. Macro Regional Proxy Mapping

C. The Numerical Dominance Problem

We observed that the distance feature (*Distance_km*) was numerically dominating the economic index (*GSCPI*). Distances often exceeded 10,000 km, while the *GSCPI* values hovered between -1 and 4.

Solution: Z score Standardization. We implemented a scaling pipeline to ensure every feature had a mean of zero ($\mu = 0$) and a standard deviation of one ($\sigma = 1$). Geometric algorithms like SVM calculate the maximum margin hyperplane based on Euclidean distance; without scaling, the model would treat a 1 km change in distance as equal to a 1 unit change in economic pressure.

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

D. Debugging the Zero Recall State

Early versions achieved 78% accuracy but 0.00 recall for disruptions.

Problem: Majority Class Bias. Because 78% of shipments were Normal, the model learned it could achieve high accuracy by guessing Normal every time.

The Solution: Cost Sensitive Weighting. We implemented `class_weight='balanced'` within the SVC. This modifies the algorithm's objective function to penalize a Missed Disruption significantly more than a False Alarm, shifting the recall from 0.00 to 0.74.

E. Handling Multi Dimensional Sparsity

Combining weather volatility including gusts and precipitation with economic drag from the *GSCPI* created a very sparse feature space. Standard linear models could not find a clear line to separate Risk from Normal because the stressors are compounding. For example, high wind only causes a delay if the port is already economically congested.

Solution: RBF Kernel Mapping. We chose the Radial Basis Function kernel to project this sparse data into a higher dimensional Hilbert space, allowing the model to identify Islands of Risk or non linear zones where high *GSCPI* and high wind speed created a disruption boundary.

IV. MACHINE LEARNING IMPLEMENTATION

A. Theoretical Selection of Architectures

We evaluated three mathematical paradigms:

- **SVM:** Chosen for robust decision boundaries in high dimensional spaces.
- **MLP:** Chosen to test deep hierarchical feature extraction.
- **Random Forest:** Chosen as a baseline for feature importance.

B. Support Vector Machines (SVM): Geometric Precision

We implemented an RBF kernel to draw a complex decision boundary around disruption clusters. Traditional linear classifiers fail on supply chain data because risks are conditional for example bad weather only causes delays if economic pressure is already high. To resolve this, we utilize the Radial Basis Function (RBF) kernel to project our input vectors $\mathbf{x}$ into an infinite dimensional feature space. The RBF kernel computes the similarity between two supply chain states, $\mathbf{x}_i$ and $\mathbf{x}_j$, as a decaying function of their squared Euclidean distance:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$$

This mathematical projection is critical. It allows the SVM to construct a non linear decision boundary capable of isolating Islands of Risk where specific, complex combinations of high *GSCPI* and localized weather events intersect. We utilized `GridSearchCV` to optimize C and γ , settling on a moderate C to allow for a soft margin that ignores minor noise in the national weather proxies.

C. Multi Layer Perceptron (MLP): Deep Learning Challenges

We settled on a 3 layer architecture with 100, 50, and 25 neurons. Early trials used the Sigmoid activation function, which led to stagnant learning. During early trials, the MLP network suffered from vanishing gradients when utilizing the standard Sigmoid activation function, $\sigma(x) = \frac{1}{1+e^{-x}}$. Because the derivative of the Sigmoid function, $\sigma'(x) = \sigma(x)(1 - \sigma(x))$, has a maximum value of 0.25, the gradients shrank exponentially during backpropagation across the hidden layers, preventing the network from learning the minority disruption class.

To preserve gradient flow, we replaced it with the Rectified Linear Unit (ReLU) activation function:

$$f(x) = \max(0, x)$$

Because the derivative of ReLU is exactly 1 for all positive inputs, it effectively solved the vanishing gradient issue, allowing the Adam optimizer to continuously adjust weights based on the sparse meteorological signals. However, the MLP struggled with the majority class bias even with the Adam optimizer.

D. Random Forest: The Interpretability Baseline

We utilized 100 decision trees. We deliberately did not prune the depth initially to allow the model to capture minute interactions between *GSCPI* and gusts. This model confirmed that *Distance_km* and *GSCPI* were the dominant predictors.

E. The Engineering Breakthrough: Cost Sensitive Objective Functions

We modified the Objective Function to treat a single disruption as being worth approximately 3.5 times more than a normal shipment in the loss calculation. To solve the Accuracy Recall Paradox and mitigate majority class bias, we mathematically altered the SVM's optimization objective.

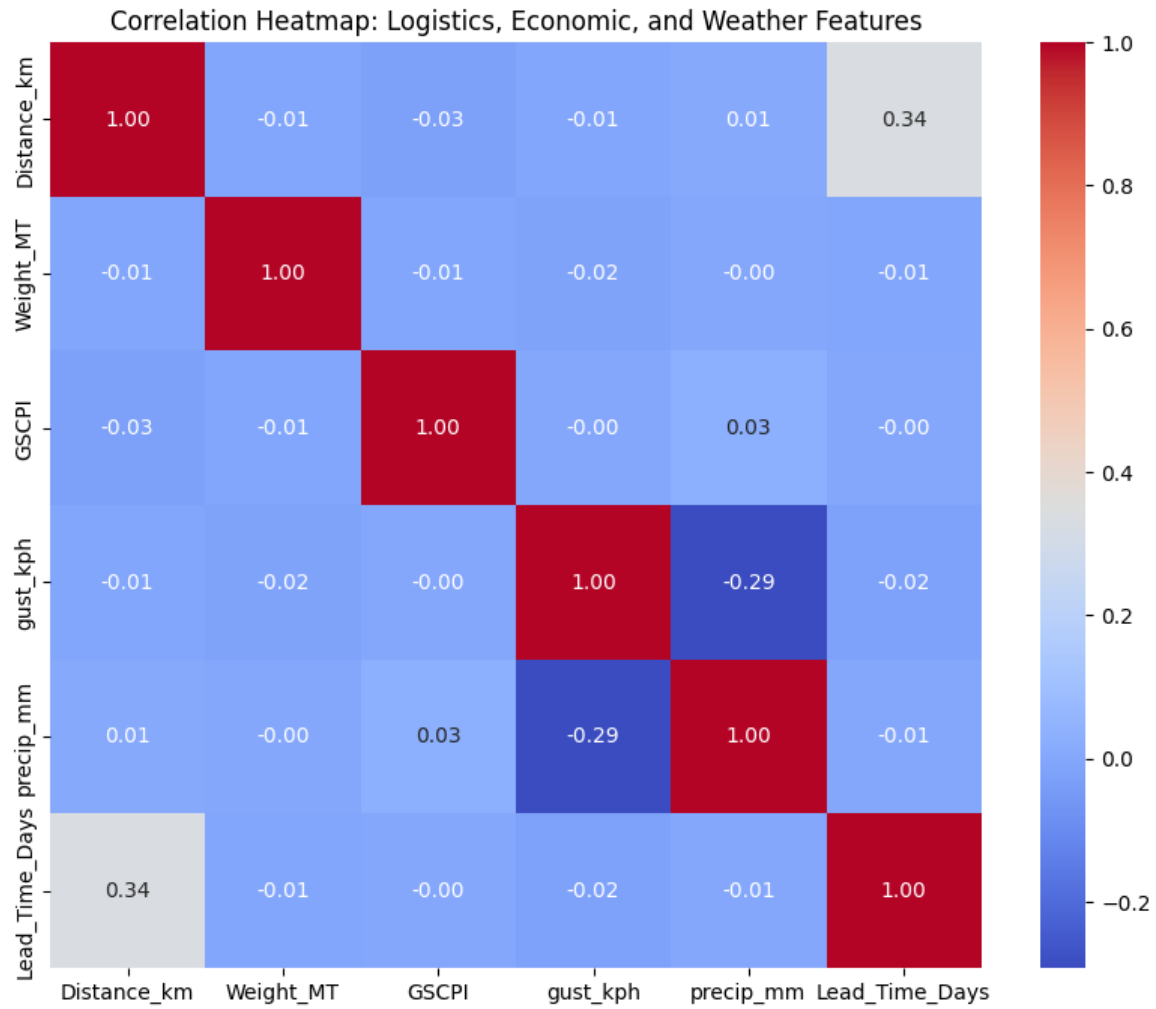


Fig. 1. Correlation Heatmap: Visualizing the statistical relationships between fused logistics logs, macroeconomic indices (GSCPI), and meteorological stressors. Note the correlation between Distance and Lead Time compared to stochastic weather noise.

In a standard soft margin SVM, the algorithm minimizes the margin vector $\mathbf{w}$ and the slack variables ξ_i . By implementing cost sensitive weighting, we introduced a class specific penalty weight, W_i , into the objective function:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n W_i \xi_i$$

By assigning a significantly higher penalty weight (W_i) to the minority Disruption class, we forced the optimization algorithm to accept a narrower overall margin in exchange for correctly classifying the costly disruption events, directly aligning the model's mathematical objective with our industrial risk management goals.

```
# The Engineering Fix: Cost Sensitive Weighting
# This shifted the recall from 0.00 to 0.74
from sklearn.svm import SVC
svm_model = SVC(kernel='rbf', class_weight='balanced', random_state=42)
svm_model.fit(X_train_scaled, y_train)
```

Listing 2. Optimized Model Logic

V. COMPARATIVE ANALYSIS AND DISCUSSION

A. The Accuracy Recall Paradox

Nominal accuracy was found to be a deceptive indicator of model utility. The MLP and Random Forest achieved nominal accuracy scores of 72.60% and 71.10% respectively, but yielded recalls of only 0.20 and 0.18. Detailed analysis revealed these architectures achieved high scores by essentially defaulting to the majority Normal class or 78% of the dataset.

B. Superiority of the Balanced SVM

The Balanced SVM identifies nearly three quarters of all actual disruption risks. In global logistics, the cost of a False Negative is exponentially higher than a False Positive.

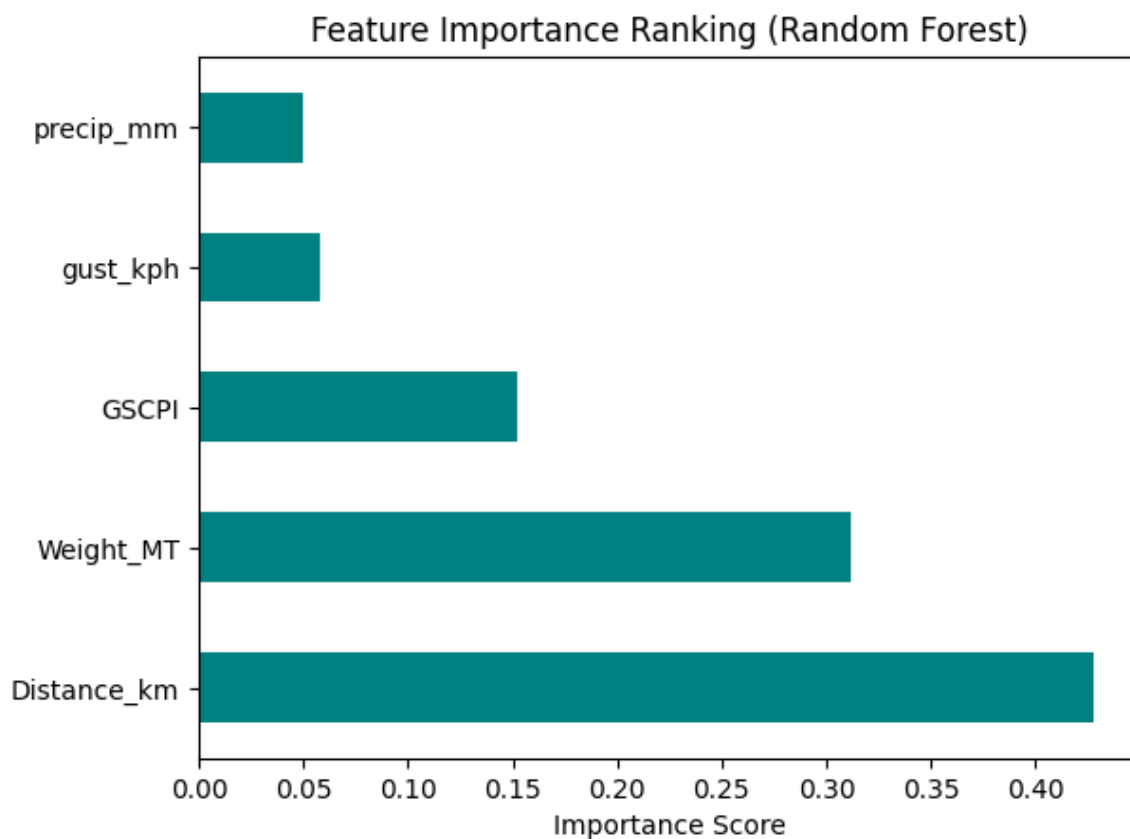


Fig. 2. Feature Importance Ranking (Random Forest): Ranking the predictive weight of fused variables, identifying Distance and Weight as dominant features while Weather features act as stochastic triggers.

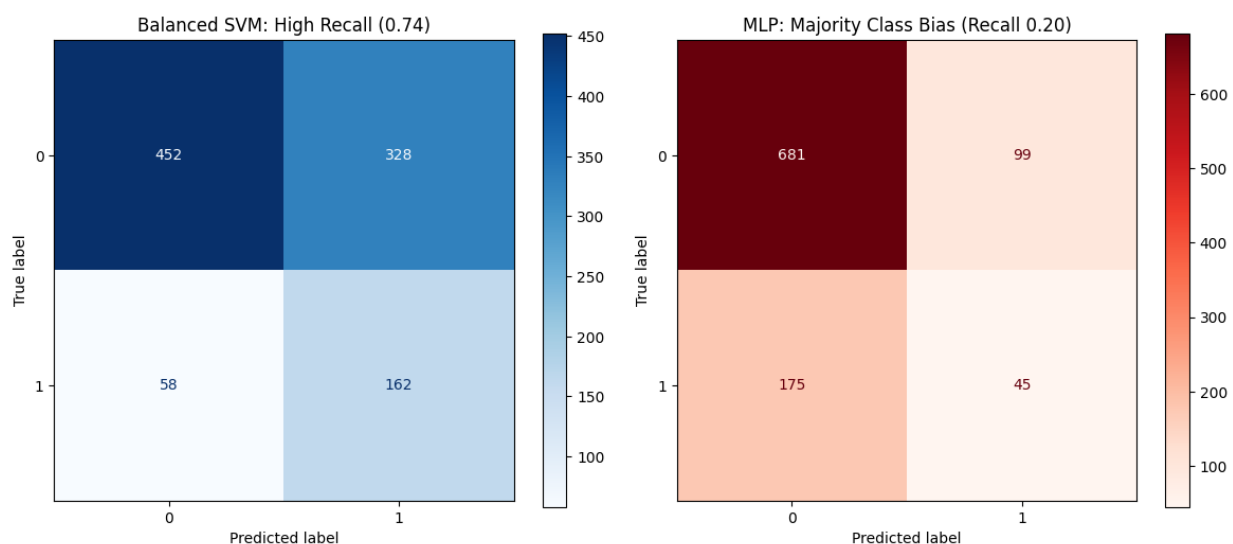


Fig. 3. Confusion Matrix Comparison: The Balanced SVM (Left) successfully navigates class imbalance to achieve high recall (0.74), whereas the MLP (Right) exhibits severe majority class bias, failing to detect the minority disruption class (Recall 0.20). This visualization serves as primary evidence of the Accuracy Recall Paradox.

C. Algorithmic Latency and Efficiency

The Balanced SVM proved most efficient with a latency of 0.276s, compared to the MLP's 3.295s. For rapid response logistics where signals update frequently, the low latency profile of the SVM is superior.

TABLE I
COMPARATIVE MODEL PERFORMANCE METRICS

Model	Accuracy	Recall (Risk)	Latency (s)
Balanced SVM	61.40%	0.74	0.276
MLP Network	72.60%	0.20	3.295
Random Forest	71.10%	0.18	0.450

D. Feature Importance and Correlation Synthesis

As visualized in the preceding sections, the analysis identified **Distance_km** and **Weight_MT** as primary drivers, followed by **GSCPI**. The Correlation Heatmap and Random Forest rankings successfully illustrated a significant relationship between systemic pressure and logistical disruptions.

VI. FUTURE DIRECTIONS

A. Advanced Class Balancing and Temporal Sequences

We propose implementing SMOTE or ADASYN sampling to build more granular decision boundaries. Furthermore, transitioning to LSTM or GRU architectures would allow the model to capture the temporal momentum of risk.

VII. CONCLUSION

A. Synthesis of the Multi Domain Framework

This research confirmed that lead time disruptions are the result of compounding pressures. The quality of the data fusion layer such as the Macro Regional Proxy mapping and Z score standardization was more influential than the neural network's complexity.

B. Final Verdict on the Accuracy Recall Paradox

We rigorously documented how a model can appear successful on paper while being a catastrophic failure in practice. The 0.74 recall score of the Balanced SVM stands as a benchmark for high stakes, imbalanced environments, providing a blueprint for Prescriptive Resilience.

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